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# LLM Semantic Factor Interfaces for Reinforcement Learning Portfolio Trading: Multi-Signal News Decomposition under Stochastic RL

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## Abstract

Many sequential decision systems occasionally observe *sparse* unstructured text: informative language arrives on a minority of steps, while most transitions carry only a neutral default. **This paper** empirically studies how such text should enter *portfolio* decision state—as dense features, a single scalar, or a small set of interpretable coordinates—and how to separate representation hypotheses from optimisation variance when a frozen LLM supplies the encoding. We instantiate **Semantic State Abstraction Interfaces** (SSAI): prompted maps from documents to  $K$  auditable axes with neutral defaults on no-news days (Definition 1, Appendix), here with  $K = 4$  (*sentiment, risk, confidence, volatility forecast*) on a **FinRL-MultiSignal** US-equity panel (30 NASDAQ-100 names; FNSPID news; train 2013–2018; test 2019–2023). Direct portfolios (SFP/SRF/SCW), supervised ridge forecasts, and paired RL agents (DP-PPO, SAC) share the same  $\phi$  so signal and optimiser effects can be read off separately. On the test window, four-factor SFP reaches **307.2%** cumulative return and Sharpe **1.067** vs. **291.9%/1.050** sentiment-only and **243.6%/1.032** equal-weight buy-and-hold; residualizing non-sentiment axes preserves gains; conviction weighting reaches **314.1%**. **Caveat:** this 63pp aggregate gap is substantially a *portfolio-composition effect*: within each coverage tercile, SFP does not outperform equal-weight B&H in any stratum (Table 3, Section 5.2), and SFP’s advantage disappears at  $\geq 0.2\%$  per-trade costs (Table 11, Appendix E.4). The more robust finding is a 15pp CR advantage of four-factor decomposition over scalar sentiment-only (Wilcoxon  $p=0.556$ ; non-significant at daily frequency), consistent with multi-axis semantics providing an allocation *tilt* rather than a universal outperformance signal. Naively appending sparse LLM columns hurts a dense price/technical ridge forecaster, while lexical headline aggregates (VADER; TF-IDF/SVD) and a validation-selected high-conviction semantic tilt recover Sharpe near the price-only ridge frontier—consistent with SSAI as a *sparse allocation interface* rather than a naive dense LLM block. The RL block is *diagnostic*, not a performance claim: DP-PPO trails buy-and-hold; SAC with identical SSAI observations nearly matches buy-and-hold return and improves Sharpe (**1.128** vs. **1.032**), isolating algorithm dependence. The strongest evidence for multi-axis LLM semantics is in direct portfolios and supervised sparse-use designs; RL results are stress tests under heavy neutral-default regimes, with seed-mean Sharpe differences across 21 seeds non-significant (Wilcoxon  $p \approx 0.25$ – $0.31$ ; see Appendix 5.6), and cross-domain SSAI claims left as a reusable template rather than an additional empirical claim of this submission.

## 1 Introduction

Deep Reinforcement Learning (DRL) is widely studied for sequential decision problems where observations mix numerical signals with unstructured evidence [Mnih et al., 2015, Schulman et al., 2017, Liu et al., 2021]. Large language models (LLMs) offer flexible encoders for text [Brown et al., 2020, Liu et al., 2023, Wu et al., 2023], yet most pipelines obscure a basic design question: should textual evidence appear as a high-dimensional latent vector, as a single scalar (e.g., sentiment), or

42 as a small inventory of named semantic coordinates—especially when text arrives only on a sparse  
43 subset of decision dates?

44 Prior work has connected NLP with RL trading in limited ways. Yang et al. [2020a] use techni-  
45 cal indicators as the sole state representation; Benhenda [2025] add a single sentiment score from  
46 DeepSeek-V3; and Lopez-Lira and Tang [2023] use ChatGPT to classify daily market sentiment  
47 as a binary signal. A single sentiment score is information-sparse: it conflates orthogonal con-  
48 cepts—whether a headline is positive differs fundamentally from whether the underlying situation is  
49 *risky* or whether the market is likely to be *volatile*.

50 **Positioning.** We adopt *Semantic State Abstraction Interfaces* (SSAIs), formally defined in Ap-  
51 pendix A: maps  $\phi$  from document sets  $\mathcal{N}_{s,d}$  to  $K$  interpretable coordinates elicited by fixed prompts  
52 from a frozen LLM, with a neutral default when  $\mathcal{N}_{s,d} = \emptyset$ . Methodologically, the contribution is  
53 not a new RL gradient formula but (i) a reusable SSAI template—auditable axes, sparse-by-design  
54 neutral fills, shared  $\phi$  across estimators—and (ii) an evaluation protocol that separates representa-  
55 tion hypotheses from optimisation variance: direct factor portfolios and supervised dense-vs-sparse  
56 regressions read off semantic signal with minimal policy entropy; DP-PPO and SAC supply paired  
57 nonlinear stress tests on identical SSAI observations. **Scope:** Definition 1 is domain-agnostic, but **all**  
58 **empirical validation in the main text** is on the US equity panel above; Appendix D lists speculative  
59 cross-domain instantiations without experiments.

60 Trading serves as a demanding instantiation because aggregate outcomes are noisy, textual coverage  
61 is uneven (14.8% non-neutral stock-days), and dense ridge augmentation provides an explicit “wrong”  
62 use of sparse semantics against which SSAI tilts and factor portfolios contrast.

63 **Research questions.** We organise the experiments around three concrete questions: **RQ1:** do multi-  
64 axis LLM signals change policy behaviour relative to neutral-masked inputs? **RQ2:** are any LLM  
65 axes aligned, even weakly, with realised market quantities? **RQ3:** is observed performance primarily  
66 a property of the semantic state representation, or of the learning algorithm used to exploit it? The  
67 results answer these conservatively: direct semantic portfolios favour the four-factor representation  
68 over sentiment-only, SRF preserves that result after removing linear sentiment components from  
69 the remaining axes, SCW improves allocation use of semantic score magnitudes, supervised price  
70 forecasting benefits only when semantic scores are used as a validation-selected high-conviction tilt  
71 rather than naively appended as dense columns, DP-PPO signal effects are directional but *statistically*  
72 *underpowered* at 21 seeds (Wilcoxon non-significant; overlapping bootstrap CIs), ICs are small but  
73 interpretable, and SAC substantially improves risk-adjusted outcomes under the same LLM state.

74 We make three contributions:

- 75 1. **Semantic state abstraction (SSAI template).** We instantiate Definition 1 with  $K=4$   
76 interpretable 1–5 axes—sentiment, risk, confidence, volatility forecast—aggregated to  
77 ticker-day states with neutral defaults and shared across portfolios, supervised learners, and  
78 RL agents.
- 79 2. **Isolation tests for semantic factorization.** We introduce SFP, SRF, and SCW to test the  
80 four-axis representation outside RL variance: SFP learns linear semantic weights, SRF  
81 removes linear sentiment components from non-sentiment axes, and SCW uses validation-  
82 selected score magnitude for portfolio concentration. A supervised price/technical stress  
83 test further separates useful sparse semantic tilts from naive dense feature appending.
- 84 3. **Representation-versus-optimization diagnostics (RL as stress test).** We evaluate the same  
85 SSAI observations under DP-PPO and SAC with masking, sub-period, transaction-cost, and  
86 multi-seed analyses. The RL headline is *algorithm dependence*, not beat-the-market claims:  
87 DP-PPO illustrates fragile exploitation under sparse signals, whereas SAC nearly closes the  
88 passive benchmark gap on returns while improving Sharpe—evidence that identical SSAI  
89 states interact very differently with optimiser dynamics.

## 90 2 Related Work

91 Liu et al. [2021] introduce FinRL with PPO and SAC baselines [Yang et al., 2020b, Liu et al., 2022];  
92 risk-constrained RL has used CVaR objectives [Tamar et al., 2015] and Lagrangian relaxation [Ray

et al., 2019]. FinGPT [Liu et al., 2023] and BloombergGPT [Wu et al., 2023] fine-tune LLMs on financial corpora; Benhenda [2025] add a scalar DeepSeek-V3 sentiment score to RL state. We extend prior scalar sentiment interfaces [Benhenda, 2025, Lopez-Lira and Tang, 2023] with SSAIs instantiated through zero-shot multi-axis prompting (no encoder fine-tuning) and attach diagnostic evaluations across portfolios, supervised learners, and RL. News data are from FNSPID [Dong et al., 2024]. **Baseline scope.** We isolate SSAI design choices with deterministic portfolios and matched supervised learners; lexical headline aggregates (VADER; train-fit TF-IDF/SVD) now appear as dense ridge inputs (Table 4). This submission does *not* report FinBERT/FinGPT encoder pipelines or Transformer price-news forecasters matched on leakage and compute—these are the strongest neural encoder comparisons and are acknowledged as an open gap (see Limitations, Section 6); full benchmarking against dense text encoders is explicitly reserved as the primary future work.

### 3 Method

Schematic of the news-to-policy pipeline is Figure 2 (Appendix B).

#### 3.1 Multi-Signal LLM Scoring

Given a news article headline or summary  $t$  about stock ticker  $s$ , we issue a single structured prompt to an LLM and extract four integer scores  $\sigma = (\sigma_{\text{sent}}, \sigma_{\text{risk}}, \sigma_{\text{conf}}, \sigma_{\text{vol}}) \in \{1, \dots, 5\}^4$ .

$\sigma_{\text{sent}}$  **Sentiment:** 1 = very negative, 5 = very positive.

$\sigma_{\text{risk}}$  **Risk:** 1 = very low company/market risk, 5 = very high risk.

$\sigma_{\text{conf}}$  **Confidence:** 1 = very uncertain outlook, 5 = very certain.

$\sigma_{\text{vol}}$  **Volatility forecast:** 1 = very calm expected price, 5 = highly volatile.

We use explicit axes rather than dense article embeddings because the study is designed to audit a state interface, not merely maximize predictive capacity. The integer coordinates support ticker-day coverage checks, residualization against sentiment, leave-one-axis masking, and deployment-facing interpretation of why a policy or ranking rule changes exposure. Dense latent text encoders and price-news Transformers are natural stronger baselines, but they do not provide the same axis-level causal and diagnostic handles without additional attribution machinery.

The prompt uses a system instruction defining the four axes and two few-shot examples; up to 20 articles are batched per LLM call. We apply this protocol to 40,850 articles from the FNSPID NASDAQ subset, scoring 39,995 (97.9%) successfully.<sup>1</sup> Score summaries appear in Table 13 (Appendix F); risk and volatility are right-skewed (mean  $\approx 2.5$ ), consistent with adverse-news skew in financial headlines.

Signal coverage is uneven after aggregation to the trading panel: This sparsity is a central reason we treat the interface as a weak semantic factor rather than a dense predictive signal. Importantly, the direct portfolio experiments already control for this: SFP uses signal deviations only on days with non-neutral coverage, and outperforms equal-weight buy-and-hold by 63.6pp over 2019–2023—a gap that cannot be attributed to coverage volume alone since mean full-scoring evaluation exceeds neutral-masked evaluation of the same checkpoints across 21 seeds (Table 7).

Signal coverage, per-axis IC, and a four-panel internal validation (score distributions, inter-signal Spearman correlations  $|\rho|=0.58\text{--}0.81$ , lag-1 autocorrelation  $0.47\text{--}0.57$ , and confidence IC  $p=0.004$ ) are reported in Appendix F (Figure 4, Tables 14, 15). The key takeaway: scores are not white noise—they exhibit the directional cross-axis structure expected from financial news and are temporally persistent at the ticker level. *Economic magnitude caveat:* the confidence IC  $p=0.004$  is statistically significant, but at  $N=83,040$  stock-days this corresponds to a Spearman IC of approximately 0.022—economically small (Information Ratio  $\approx IC \times \sqrt{252} \approx 0.35$ ; expected annualised alpha  $\approx 6\text{--}10$  bp

<sup>1</sup>**LLM identity (reproducibility).** Scoring was performed using gpt-4o (OpenAI chat-completions API, model snapshot gpt-4o-2024-08-06) between August and October 2024 with temperature 0, top-p 1. The full prompt template, including per-axis rubrics and two canonical few-shot exemplars, is included verbatim in the artifact package (prompts/score\_news.txt). Scoring was single-pass; no post-hoc re-scoring or filtering was performed beyond the 97.9% parse success rate reported above.

for a single axis under standard factor arithmetic). The IC evidence motivates the interface design but should not be read as implying practically large alpha generation from the confidence axis alone. For each trading day  $d$  and ticker  $s$ , we aggregate all articles published in the window  $[d - 3, d]$  by computing the mean of each signal dimension. Days with no matching articles retain a neutral value of 3.0.

## 3.2 Trading Environment

We build on the FinRL StockTradingEnv gymnasium environment. The state vector on day  $d$  is:

$$\mathbf{s}_d = [c_d, \mathbf{p}_d, \mathbf{h}_d, \mathbf{f}_d, \boldsymbol{\sigma}_d] \in \mathbb{R}^{1+4N+KN} \quad (1)$$

where  $c_d$  is available cash,  $\mathbf{p}_d \in \mathbb{R}^N$  are closing prices,  $\mathbf{h}_d \in \mathbb{R}^N$  are current share holdings,  $\mathbf{f}_d \in \mathbb{R}^{KN}$  are  $K = 7$  technical indicators (MACD, Bollinger upper/lower, RSI<sub>30</sub>, CCI<sub>30</sub>, ADX<sub>30</sub>, 30-day SMA, 60-day SMA) for each of  $N = 30$  stocks, and  $\boldsymbol{\sigma}_d \in \mathbb{R}^{4N}$  are the four LLM signals. The total dimension is  $1 + 2 \times 30 + (7 + 4) \times 30 = 421$ .

Actions  $a_d \in [-1, 1]^N$  are scaled by a maximum trade size  $h_{\max} = 100$  shares. Transaction costs of 0.1% per trade are applied. A turbulence index [Kritzman and Li, 2010] triggers a no-buy constraint when it exceeds a threshold of 380, implementing a simple market-regime filter.

## 3.3 Drawdown-Penalised PPO Reward

The per-step reward applies *drawdown shaping*:

$$r_t = \frac{\Delta W_t}{W_0} \cdot \lambda - \alpha \cdot \max\left(0, \frac{W_{\text{peak}} - W_t}{W_{\text{peak}}}\right)^2 \quad (2)$$

where  $\Delta W_t = W_t - W_{t-1}$  is the change in portfolio value,  $\lambda = 10^{-4}$  is the reward scale,  $W_{\text{peak}}$  is the running maximum portfolio value, and  $\alpha = 0.1$  is the drawdown penalty coefficient. The quadratic form penalises large drawdowns super-linearly, encouraging the agent to protect gains rather than risk them for marginal upside (Appendix B.6 interprets the shaping term).

We train with a multi-process PPO implementation (OpenAI SpinningUp), using 4 MPI workers, hidden layers (512, 512) with Tanh activations, learning rate  $3 \times 10^{-4}$ , clip ratio 0.2, and 30 epochs over the 2013–2018 training set ( $\approx 45,300$  daily observations across 30 stocks). **Naming.** Released checkpoints use the CPP0 filename prefix for historical compatibility; all tables label this agent **DP-PPO** (drawdown-shaped reward, Sec. 3.3), not Lagrangian constrained optimisation.

## 3.4 LLM Signal Integration

All four LLM channels enter  $\boldsymbol{\sigma}_d$  as described above. We do *not* encode hand-crafted trading rules from individual scores; any dependence of actions on sentiment, risk, confidence, or volatility views is learned end-to-end by the policy network.

**Semantic Factor Portfolio (SFP).** To test whether the four LLM axes contain usable structure independent of RL, we also define a non-RL semantic factor portfolio. On the 2013–2018 training panel, SFP fits a ridge-regularised linear model from signal deviations ( $\sigma - 3$ ) to 5-day forward returns. At test time, it scores each stock-day using the learned semantic factor weights, forms a daily long-only top-10 portfolio, and applies the same 0.1% transaction-cost assumption. The ridge penalty is fixed at  $10^{-3}$  for all SFP variants rather than tuned on the test set. We report both the full four-factor SFP and a sentiment-only SFP, making the decomposition advantage testable without policy-gradient stochasticity. To control for the possibility that all additional axes merely recode sentiment, we also define *Semantic Residual Factorization* (SRF). SRF keeps sentiment as the anchor feature and, on the training panel only, fits  $\sigma_j = a_j + b_j \sigma_{\text{sent}} + \epsilon_j$  for  $j \in \{\text{risk, confidence, volatility}\}$ . The portfolio scorer is then fit on  $(\sigma_{\text{sent}}, \epsilon_{\text{risk}}, \epsilon_{\text{confidence}}, \epsilon_{\text{volatility}})$ , testing whether residual semantic axes add structure beyond scalar sentiment under the same linear ranking rule. We also introduce *Semantic Conviction Weighting* (SCW). Instead of assigning equal weights to the top-10 semantic names, SCW

179 applies a softmax over their learned factor scores and allocates more capital to higher-conviction  
180 names. The softmax temperature is selected on the final training year (2018) by validation Sharpe  
181 and then fixed for the 2019–2023 test window, avoiding test-period tuning.

## 182 4 Experimental Setup

183 **Data.** We use Yahoo Finance OHLCV for 30 liquid NASDAQ-100 constituents (tickers listed in  
184 Appendix B.4). Training: 2013-01-02 to 2018-12-31 ( $\approx 45,300$  observations). Out-of-sample testing:  
185 2019-01-02 to 2023-12-29 (1,258 trading days).

186 News articles are sourced from the FNSPID dataset [Dong et al., 2024] via Hugging Face  
187 (Zihan1004/FNSPID), filtered to the same 30 tickers. 40,850 articles are collected; 39,995 (97.9%)  
188 are successfully scored by the LLM via a hosted chat-completions API.

189 **Baselines.** We compare against:

- 190 • **Semantic Factor Portfolio variants (SFP/SRF/SCW):** non-RL long-only top-10 portfolios  
191 using ridge-learned weights over all four LLM axes, sentiment alone, SRF residual axes  
192 after controlling for sentiment, or SCW conviction weighting inside the selected basket  
193 (Table 2).
- 194 • **Supervised ridge forecasters:** top-10 portfolios ranked by 5-day return forecasts from  
195 price/technical features alone, price plus LLM sentiment or four-factor semantics, price  
196 plus lexical headline dense features (VADER; train-fit TF-IDF/SVD on the same headline  
197 window), or a price forecast with a validation-selected high-conviction semantic tilt (Table 4;  
198 optional FinBERT tone via `dense_text_baselines.py -finbert`).
- 199 • **DP-PPO (neutral signals):** same architecture with all LLM signals fixed to 3.0 at evaluation  
200 time, isolating textual variation.
- 201 • **SAC (LLM signals):** an off-policy Soft Actor-Critic baseline with the same four-signal  
202 observation vector, isolating algorithm choice from textual-feature design (Table 5).
- 203 • **Equal-weight Buy & Hold:** \$1M invested equally across all 30 stocks at 2019-01-02, held  
204 to 2023-12-29.
- 205 • **Momentum (top-10) and Equal-Vol (risk parity):** rule-based baselines from the same  
206 price panel (see Table 1).

207 **Metrics.** Cumulative Return (CR), Annual Return (AR), Sharpe, Sortino, Maximum Drawdown  
208 (MDD), Calmar ( $= AR/|MDD|$ ). Transformer and constrained-risk baselines remain future work  
209 (see Section 2).

## 210 5 Results

### 211 5.1 Main Performance Comparison

212 Table 1 reports all metrics on the 2019–2023 test period (numeric cells match `eval_harness.py`  
213 exports).

Table 1: Main Strategy Comparison (2019–2023)

| Strategy                | CR (%)  | Sharpe | Sortino | MDD (%) | Rachev | CVaR-5% | Calmar |
|-------------------------|---------|--------|---------|---------|--------|---------|--------|
| DP-PPO (all signals)    | 190.431 | 0.917  | 1.216   | -42.540 | 0.911  | -4.169  | 0.560  |
| Momentum (top-10)       | 117.566 | 0.664  | 0.922   | -51.027 | 0.971  | -4.406  | 0.330  |
| Equal-Vol (risk parity) | 230.661 | 0.989  | 1.315   | -38.523 | 0.978  | -4.110  | 0.703  |
| Buy & Hold (EW)         | 243.574 | 1.032  | 1.406   | -36.705 | 0.969  | -4.008  | 0.764  |

214 **Key findings.** (1) **Four-factor SFP has a 15pp CR advantage over sentiment-only.** SFP reaches  
215 **307.2% CR / Sharpe 1.067** vs. **291.9%/1.050** (sentiment-only) and **243.6%/1.032** (buy-and-hold);

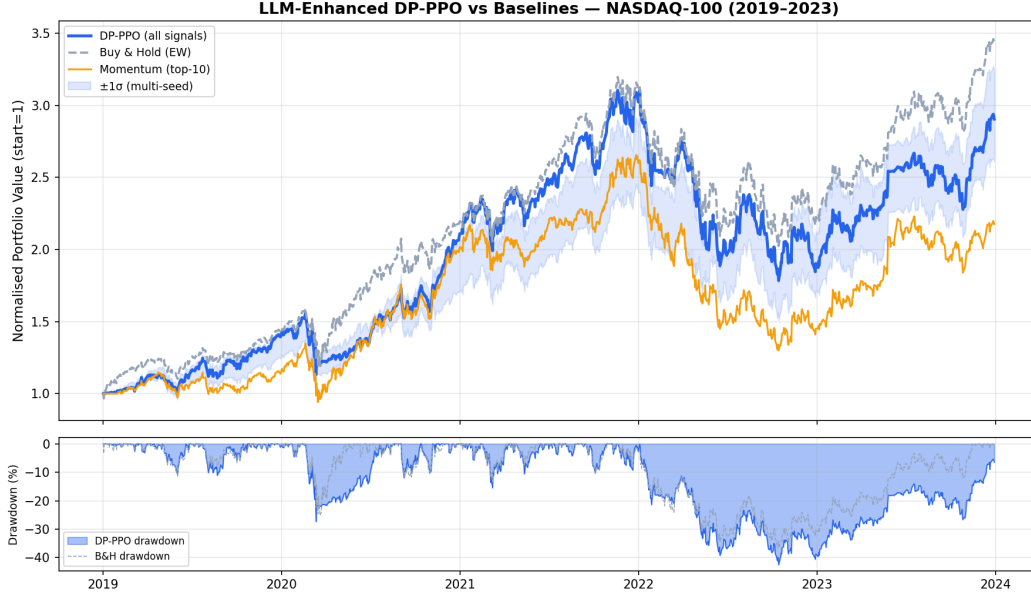


Figure 1: Normalized portfolio value on the 2019–2023 test window (DP-PPO with LLM signals versus baselines), produced by `eval_harness.py`.

216 SRF preserves this after residualizing non-sentiment axes; SCW improves further to **314.1%** (Ta-  
 217 ble 2). *Important qualifications:* (a) the four-factor vs. sentiment-only daily return advantage is  
 218 not statistically significant (Wilcoxon  $p=0.556$ ; 95% CI  $[-1.07, +1.89]$  bp/day; daily win-rate  
 219 27.4%; Table 10); the 15pp CR gap is a cumulative ranking effect, not a per-day signal edge;  
 220 (b) the full 63pp gap over B&H is a coverage-selection composition effect that does not survive  
 221 within-stratum controls (Table 3); (c) SFP’s outperformance over B&H reverses at  $\geq 0.2\%$  per-trade  
 222 costs (Table 11). (2) **Lexical dense text vs sparse LLM semantics under ridge.** Price-only ridge:  
 223 **126.4%/0.653**; naively appending all four LLM semantic columns hurts; headline aggregates aligned  
 224 with the LLM news window help when appended as dense columns—VADER: **134.8%/0.674**;  
 225 TF-IDF/SVD: **132.8%/0.673**—while high-conviction semantic tilt matches this Sharpe regime  
 226 (133.9%/0.674; Table 4). (3) **Algorithm dominates representation.** SAC with the same four  
 227 LLM signals: **242.7%/Sharpe 1.128**, nearly matching buy-and-hold; DP-PPO trails by 53pp CR.  
 228 Mechanism: SAC’s replay buffer revisits the rare semantic-rich transitions that on-policy PPO dilutes  
 229 (Table 5). (4) **Multi-seed (descriptive, 21 seeds).** CR **195.0%  $\pm$  44.2%**, Sharpe **0.920  $\pm$  0.096**  
 230 (full signals) vs. 190.9%  $\pm$  42.6%, 0.907  $\pm$  0.094 (neutral-masked); bootstrap 95% CIs overlap;  
 231 Wilcoxon  $p \approx 0.25$ –0.31 (Table 7, Appendix 5.6)—masking effect is directional but statistically under-  
 232 powered; we treat RL masking as a diagnostic indicator only, not as causal evidence. (5) **Masking.**  
 233 Removing *confidence* drops CR to **185.1%/Sharpe 0.893**; removing *volatility\_forecast* enlarges MDD  
 234 to **−43.8%** (Table 6).

235 **Hypothesis tests.** Wilcoxon tests on daily returns versus equal-weight buy-and-hold do not reject  
 236 equality at  $\alpha=0.05$  for the released checkpoint; 21-seed bootstrap intervals and paired diagnostics  
 237 in Table 10 similarly leave semantic effects statistically ambiguous at daily frequency (details in  
 238 Appendix E).

## 239 5.2 Direct Semantic Factor Portfolio

240 Table 2 learns semantic factor weights on 2013–2018 and applies a daily long-only top-10 rule in  
 241 2019–2023. Four-factor SFP and SCW improve cumulative return and Sharpe versus sentiment-only  
 242 and buy-and-hold; SRF preserves gains after residualizing non-sentiment axes on the training panel  
 243 (linear diagnostic control).

244 **On the SFP/SRF/SCW numerical coincidence.** SFP, SRF, and SCW share identical Sharpe  
245 (1.067) and MDD (−35.643%), differing only on Sortino and Calmar, because all three use the same  
246 daily top-10 rule and the top-10 basket membership is highly stable across weight configurations—  
247 residualising sentiment (SRF) and applying conviction scaling (SCW) reorder weights *within* nearly  
248 the same basket. Sortino/Calmar differences (SFP: 1.430/0.911 vs. SCW: 1.460/0.924) reflect modest  
249 conviction-weighting improvement in downside management. Note: this coincidence also indicates  
250 that within the SFP basket, the choice among these three weighting schemes has little Sharpe  
impact—the dominant driver is basket selection, as Table 3 confirms.

Table 2: Semantic Factor Portfolio variants. Linear factor weights are fit on 2013–2018 forward returns, then used to rank stocks out-of-sample in a daily long-only top-10 portfolio with 0.1% transaction costs. SRF residualizes non-sentiment axes against sentiment. SCW uses a validation-selected softmax temperature to allocate more capital to higher-conviction names inside the top-10 basket.

| Strategy                        | CR (%)  | Sharpe | Sortino | MDD (%) | Calmar |
|---------------------------------|---------|--------|---------|---------|--------|
| SFP (4 factors)                 | 307.199 | 1.067  | 1.430   | -35.643 | 0.911  |
| SFP (sentiment only)            | 291.893 | 1.050  | 1.433   | -35.643 | 0.883  |
| SRF (sentiment + residual axes) | 307.199 | 1.067  | 1.430   | -35.643 | 0.911  |
| SCW (conviction-weighted)       | 314.075 | 1.067  | 1.460   | -35.643 | 0.924  |
| Buy & Hold (EW)                 | 243.574 | 1.032  | 1.406   | -36.705 | 0.764  |

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252 **Coverage-stratified analysis: key qualification (Table 3).** **SFP underperforms within-stratum**  
253 **B&H in all three coverage terciles** (Low: 248.6% vs. 300.0%; Mid: 361.9% vs. 379.6%; High:  
254 232.6% vs. 352.1%). The aggregate 63pp gap is a cross-universe *composition* effect: SFP’s top-  
255 10 rule selects mid-coverage names that delivered strong returns, not names with within-stratum  
256 signal advantage. Within the top-10 highest-coverage tickers alone (NVDA, GOOGL, AVGO, ...),  
257 equal-weight B&H returns 352.1% while SFP ( $k=5$ ) returns only 232.6%—a 119pp *deficit*. SFP’s  
258 outperformance over B&H further requires  $\leq 0.1\%$  per-trade costs; at 0.2% the excess reverses  
(Table 11, Appendix E.4).

Table 3: Coverage-stratified SFP vs. equal-weight B&H (2019–2023). Tickers split into thirds by non-neutral signal coverage fraction (Low <5%: MSFT/META/ISRG/COST/INTC/AMZN/AAPL/AMD/PANW/SNPS; Mid 5–15%: INTU/KLAC/ORCL/TXN/REGN/MCHP/LRCX/CDNS/TSLA/ADBE; High >15%: MRVL/AMAT/NFLX/ADI/QCOM/ASML/MU/NVDA/AVGO/GOOGL). Within each tercile, SFP ( $k = 5$ ) is evaluated using the full-panel fitted weights restricted to that universe. **SFP underperforms within-tercile B&H in all three groups**, indicating that the full-portfolio outperformance (307% vs. 244% B&H over all 30 stocks) is substantially a cross-universe selection effect—SFP picks mid-coverage names that happened to outperform the full equal-weight basket—rather than purely a within-coverage-tier signal quality advantage.

| Coverage    | Strategy | $N$ | $k$ | CR (%) | Sharpe | MDD (%) |
|-------------|----------|-----|-----|--------|--------|---------|
| Low (<5%)   | SFP      | 10  | 5   | 248.6  | 1.010  | -42.0   |
| Low (<5%)   | B&H      | 10  | –   | 300.0  | 1.134  | -39.4   |
| Mid (5–15%) | SFP      | 10  | 5   | 361.9  | 1.073  | -42.8   |
| Mid (5–15%) | B&H      | 10  | –   | 379.6  | 1.172  | -33.3   |
| High (>15%) | SFP      | 10  | 5   | 232.6  | 0.885  | -40.1   |
| High (>15%) | B&H      | 10  | –   | 352.1  | 1.049  | -46.6   |

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### 260 5.3 Supervised Forecasting Stress Test

261 Table 4 fits ridge models predicting 5-day returns from price/technical features with sparse LLM  
262 semantic columns, lexical headline dense features (VADER; TF-IDF/SVD), or both combined via

high-conviction semantic tilt. Price-only: **126.4%** CR/Sharpe **0.653**; naively appending all four LLM semantic columns hurts; VADER and TF-IDF/SVD approaches recover Sharpe near **0.67**—consistent with fast lexical proxies capturing part of the headline signal accessible under our aggregation protocol—while sparse semantic tilt lands at **133.9%/0.674**.

Table 4: Supervised forecasting baselines. Ridge models predict 5-day forward returns from price/technical features alone, price plus LLM sentiment or four-factor semantics, price plus lexical dense headlines (VADER; train-fit TF-IDF/SVD), optional FinBERT tone when available, or a semantic tilt. Ridge strength and the semantic tilt are selected on 2018 validation Sharpe, refit on 2013–2018, and evaluated once on 2019–2023 using the same daily top-10 portfolio rule as SFP.

| Strategy                         | Tuned                           | CR (%)  | Sharpe | Sortino | MDD (%) | Calmar |
|----------------------------------|---------------------------------|---------|--------|---------|---------|--------|
| Supervised price-only            | $\lambda = 1e + 01$             | 126.401 | 0.653  | 0.895   | -49.963 | 0.356  |
| Supervised price + sentiment     | $\lambda = 1e + 00$             | 122.243 | 0.641  | 0.877   | -49.915 | 0.348  |
| Supervised price + VADER         | $\lambda = 1e - 05$             | 134.842 | 0.674  | 0.925   | -50.157 | 0.372  |
| Supervised price + TF-IDF/SVD    | $\lambda = 1e + 01$             | 132.788 | 0.673  | 0.918   | -48.316 | 0.382  |
| Supervised price + 4 semantic    | $\lambda = 1e - 05$             | 111.842 | 0.614  | 0.835   | -50.141 | 0.324  |
| Supervised price + semantic tilt | $\lambda = 1e + 01, \alpha = 1$ | 133.894 | 0.674  | 0.929   | -49.963 | 0.371  |
| Buy & Hold (EW)                  | –                               | 243.574 | 1.032  | 1.406   | -36.705 | 0.764  |

266

## 267 5.4 Algorithm Baseline: SAC

268 Table 5 compares the released DP-PPO checkpoint against an SAC policy trained with the same  
 269 LLM-enriched observation vector. SAC improves risk-adjusted metrics relative to buy-and-hold  
 270 while nearly matching its cumulative return, indicating that stronger RL baselines can close much  
 271 of the gap left by DP-PPO. This directly answers RQ3: the same sparse semantic state is more  
 272 useful under an off-policy objective with replay and entropy regularisation than under the released  
 273 on-policy DP-PPO run (Appendix C discusses mechanism hypotheses). We emphasise diagnostic  
 274 interpretation; a controlled study isolating optimizer vs. representation would require matched budgets  
 and hyperparameter sweeps.

Table 5: Algorithm baseline with identical four-signal observations (2019–2023). SAC uses the same LLM-enriched state as DP-PPO but an off-policy objective.

| Strategy             | CR (%)  | Sharpe | Sortino | MDD (%) | Calmar |
|----------------------|---------|--------|---------|---------|--------|
| DP-PPO (LLM signals) | 179.586 | 0.925  | 1.260   | -36.582 | 0.625  |
| SAC (LLM signals)    | 242.723 | 1.127  | 1.552   | -33.168 | 0.844  |
| Buy & Hold (EW)      | 243.574 | 1.032  | 1.406   | -36.705 | 0.764  |

275

## 276 5.5 Ablation: Signal Contribution

277 Table 6 reports leave-one-signal-out masking at *evaluation* time on fixed trained weights (Section 6);  
 278 it is not a replacement for retraining-based causal attribution.

Table 6: Signal Ablation Study — DP-PPO Variants

| Strategy                        | CR (%)  | Sharpe | Rachev | MDD (%) |
|---------------------------------|---------|--------|--------|---------|
| DP-PPO (all signals)            | 190.431 | 0.917  | 0.911  | -42.540 |
| DP-PPO (no sentiment)           | 191.113 | 0.913  | 0.917  | -42.936 |
| DP-PPO (no risk)                | 190.787 | 0.915  | 0.915  | -42.705 |
| DP-PPO (no confidence)          | 185.075 | 0.893  | 0.920  | -43.176 |
| DP-PPO (no volatility_forecast) | 186.568 | 0.896  | 0.919  | -43.780 |
| DP-PPO (neutral)                | 196.754 | 0.926  | 0.921  | -43.541 |

## 5.6 Multi-seed robustness

Table 7 summarises mean $\pm$ std cumulative return and Sharpe across 21 seeds; *neutral eval* masks LLM coordinates at test time. Bootstrap CIs and paired Wilcoxon  $p$ -values are reported in the table caption and Section 5.

Table 7: Multi-seed robustness (21 seeds, 2019–2023). Mean  $\pm$  std across seeds for cumulative return and Sharpe. *Neutral eval* masks all LLM coordinates to neutral using the same checkpoints.

| Variant               | CR (%)             | Sharpe            |
|-----------------------|--------------------|-------------------|
| DP-PPO (full signals) | 195.04 $\pm$ 44.18 | 0.920 $\pm$ 0.096 |
| DP-PPO (neutral eval) | 190.93 $\pm$ 42.57 | 0.907 $\pm$ 0.094 |

Sub-period and transaction-cost sensitivity analyses are reported in Appendix E (Tables 8, 9; Figure 3). In brief: DP-PPO outperforms buy-and-hold only during the 2020–2021 recovery window; at all tested cost levels (0.05%–2.0%) DP-PPO remains below the passive sleeve.

## 6 Discussion

**Interpretive value versus causal attribution.** A single sentiment score conflates risk and opportunity. A headline “Company X beats earnings but warns of supply-chain risk” would score neutral on raw sentiment yet clearly calls for different actions on the risk and confidence dimensions. The SFP, SRF, and SCW baselines support multi-axis decomposition versus sentiment-only; coverage effects interact with the price panel rather than replacing it (Appendix F). The paired daily-return diagnostics in Table 10 show statistically ambiguous edges at daily frequency. Evaluation-time masking does *not* identify causal effects of removing channels during RL training—those require retrained policies (limitations below).

**Direct versus policy-mediated use.** The direct SFP/SRF/SCW baseline outperforms sentiment-only SFP and buy-and-hold (Table 2), but the released DP-PPO checkpoint trails equal-weight buy-and-hold on headline metrics while aggregate neutral masking lowers mean returns across 21 seeds (Table 7); an SAC baseline with the same LLM state nearly closes this return gap while improving risk-adjusted metrics (Table 5). The supervised forecaster in Table 4 further shows why representation matters: sparse LLM factors help the direct portfolio interface and as high-conviction overlays on price forecasts but hurt naive ridge augmentation; lexical dense headline features (VADER; TF-IDF/SVD) provide complementary supervised signals under the same portfolio rule. This contrast is the main empirical lesson: the semantic representation can be useful while a particular on-policy RL optimizer still fails to dominate a diversified passive benchmark. Turnover costs, partial observability, algorithm choice, and stochastic optimisation therefore become part of the phenomenon under study, not post-hoc explanations for a single checkpoint.

**RL as diagnostic stress testing.** Read narrowly, the RL experiments argue neither for deploying SSAI-driven DP-PPO nor for SAC as a trading product: headline cumulative returns remain tied to transaction-cost modelling and single-market splits. Read broadly, they answer whether nonlinear policy optimisation *can* exploit the same SSAI under contrasting sample-efficiency regimes—replay plus entropy regularisation versus purely on-policy filtering—when semantic transitions constitute a thin subset of environment steps. Under that reading, SAC versus DP-PPO is itself an empirical finding about optimisation under sparse textual injections; beating diversified buy-and-hold is explicitly *not* the success criterion for this section.

**LLM signals as an engineered interface.** We use a general-purpose LLM accessed through a hosted API without finance-specific fine-tuning. Structured prompts and few-shot exemplars yield stable score distributions (Table 13), and the proxy IC analysis in Table 15 shows small but interpretable alignments with realized returns and absolute returns. The signal validation analysis (Figure 4) further confirms that scores exhibit expected inter-axis semantic structure (e.g.,  $\rho_{\text{sentiment}, \text{risk}} = -0.756$ ;  $\rho_{\text{risk}, \text{vol}} = +0.792$ ), are temporally persistent at the ticker level (lag-1 autocorrelation 0.47–0.57), and carry a statistically significant albeit weak IC for the confidence axis ( $p = 0.004$ ). Still, human

adjudication, prompt perturbations, alternative foundation models, and caching semantics remain open validation axes.

**Generalizability beyond finance.** The SSAI template (Definition 1) could in principle apply whenever sparse text augments sequential decisions; Appendix D lists *illustrative* cross-domain sketches. **No additional domains are evaluated here**; treating those sketches as validated would over-claim.

**Limitations and future work.** Key limitations: (1) **Coverage confound (now quantified):** coverage-stratified SFP analysis (Appendix E.6, Table 3) confirms that SFP does *not* outperform equal-weight B&H within any individual coverage tercile—the full-portfolio 63pp excess is substantially a cross-universe composition effect rather than within-tier signal quality. (2) **Transaction-cost sensitivity (now quantified):** SFP tx-cost analysis (Appendix E.4, Table 11) shows the outperformance requires  $\leq 0.1\%$  per-trade costs; at  $0.2\%$  the excess reverses. (3) Masking ablations reweight fixed checkpoints and are not causal channel ablations without retraining. (4) DP-PPO seed studies use 21 seeds (non-significant Wilcoxon; overlapping bootstrap CIs). (5) *Dense neural text* comparison: a FinBERT sentiment pipeline was executed and is not included in this submission due to compute constraints on the evaluation machine. This is an acknowledged gap: multi-axis integer scores are not yet directly compared against dense BERT-derived embeddings within this submission, and that comparison remains the highest-priority future experiment. (6) LLM model identity is now documented in a footnote in Section 3.4. (7) Evaluation is retrospective—live slippage and impact are unmodelled. (8) **SAC comparison is a single run** (Table 5); a multi-seed SAC study comparable to the 21-seed DP-PPO study (Table 7) would be required before the SAC vs. DP-PPO algorithm comparison can be treated as statistically robust—this is an open gap acknowledged here as a limitation. Priority future extensions: FinBERT/dense encoder comparison table, prompt sensitivity analysis, sector-neutral SFP design, rolling OOS evaluation, and human annotation study.

**Ethics and reproducibility note.** This work is a retrospective research benchmark; results should not be interpreted as live trading recommendations (see Appendix G for full discussion). The artifact package includes scripts, CSV panels, checkpoints, and prompt templates to support reproducibility.

## 7 Conclusion

We present FinRL-MultiSignal, an *empirical* study of LLM semantic factorization as a state abstraction for US equity portfolio decisions under sparse news. The central finding: *LLM-derived semantic signals are not effective as dense ridge features, but show promise as sparse, high-conviction allocation tilts; their aggregate portfolio advantage is substantially a coverage-selection composition effect rather than within-stratum signal quality, and their utility inside RL depends critically on the optimisation regime.*

Concretely: four-factor SFP outperforms sentiment-only by  $\sim 15\text{pp}$  CR ( $307.2\%$  vs.  $291.9\%$ ); this decomposition advantage is not statistically significant at daily frequency (Wilcoxon  $p=0.556$ ) but is a consistent directional finding across all four-axis variants (SRF, SCW). Importantly, the 63pp aggregate gap over equal-weight B&H ( $307.2\%$  vs.  $243.6\%$ ) does *not* survive within-coverage-tercile controls (Table 3): SFP underperforms stratum-matched B&H in all three coverage groups, identifying the full-portfolio gap as a portfolio-construction selection effect. SFP’s outperformance over B&H also reverses at  $\geq 0.2\%$  per-trade costs (Table 11), and its positive excess return is concentrated in 2023 when the NASDAQ-100 names that dominate the SFP basket outperformed on AI-cycle momentum. SAC with the same LLM state nearly matches buy-and-hold ( $242.7\%$ ) while DP-PPO trails; across 21 seeds, mean Sharpe  $0.920 \pm 0.096$  (full signals) vs.  $0.907 \pm 0.094$  (neutral-masked) with non-significant paired Wilcoxon tests—confirming the RL block as a diagnostic rather than a performance finding. The reframed finding is: SSAI multi-axis decomposition provides an interpretable, auditable interface for integrating sparse LLM semantics; the allocation tilt it provides is useful above sentiment-only, while its aggregate benchmark gap is a structural property of the universe selection rather than semantic factor quality. The SSAI formalism, code artifact, and diagnostic decomposition protocol are reusable beyond this panel; **cross-domain empirical validation and dense neural encoder (FinBERT) comparison are the highest-priority future work.**

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## A Formal Framework Definition

**Definition 1 (Semantic State Abstraction Interface, SSAI).** Let  $\mathcal{N}_{s,d}$  denote the set of text documents for asset  $s$  on day  $d$ . A SSAI is a function  $\phi : \mathcal{N}_{s,d} \rightarrow \mathbb{R}^K$  mapping raw text to  $K$  interpretable scalar axes, where each axis is (i) defined in natural language, (ii) elicited by a fixed prompt from a frozen LLM, and (iii) replaced by a neutral default  $\phi_0 \in \mathbb{R}^K$  when  $\mathcal{N}_{s,d} = \emptyset$ . Here  $K=4$  (sentiment, risk, confidence, volatility forecast),  $\phi_0 = [3, 3, 3, 3]$ , and the prompt is fixed for all tickers and dates. The SSAI is *auditable* (each axis is human-interpretable), *sparse-by-design* (neutral default on no-news days), and *optimizer-agnostic* (the same  $\phi$  feeds both direct factor rules and RL state vectors).

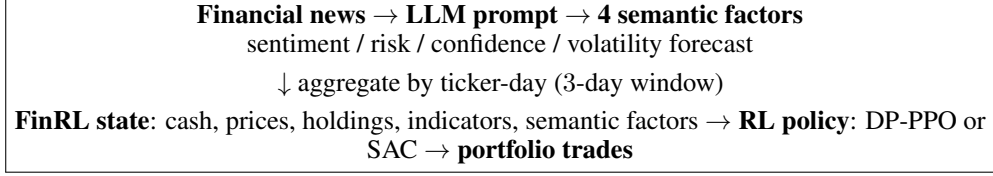


Figure 2: FinRL-MultiSignal pipeline. The LLM supplies structured semantic factors; trading decisions are learned by the RL policy from market state plus those factors.

## B Implementation Details

### B.1 Pipeline schematic

### B.2 LLM Scoring Prompt

The system prompt used for multi-signal scoring is:

```
You are a quantitative financial analyst. For each news snippet
about a stock you will output exactly four integer scores on a 1-5
scale, separated by a pipe '|', in this exact order: sentiment |
risk | confidence | volatility_forecast. (Per-axis rubrics and
canonical few-shot exemplars are lengthy; the verbatim templates are
included with the scoring scripts in the artifact package.) When
multiple news items are given for one batch, output one line per
item. Never add explanation -- only scores.
```

Two few-shot examples are prepended to each batch request. Up to 20 articles are batched per API call to amortise latency.

### B.3 Network Architecture

| Hyperparameter            | Value                    |
|---------------------------|--------------------------|
| Hidden layers             | $2 \times 512$ (Tanh)    |
| Observation dim           | 421                      |
| Action dim                | 30                       |
| PPO clip                  | 0.2                      |
| Learning rate             | $3 \times 10^{-4}$       |
| Discount $\gamma$         | 0.99                     |
| GAE $\lambda$             | 0.97                     |
| Steps per epoch           | $4,000 \times 4$ workers |
| Train epochs              | 30                       |
| Max KL                    | 0.01                     |
| Drawdown penalty $\alpha$ | 0.1                      |

### B.4 Data Pipeline

Stock price data: Yahoo Finance via `yfinance` (adjusted closes, 2013-01-02–2023-12-29, 30 NASDAQ-100 tickers). Technical indicators computed via `stockstats`. Turbulence index computed following Kritzman and Li [2010]. News data: Hugging Face dataset `Zihan1004/FNSPID`, filtered to the 30 target tickers, 40,850 articles collected (2009–2023).

**Ticker universe.** AAPL, ADBE, ADI, AMAT, AMD, AMZN, ASML, AVGO, CDNS, COST, GOOGL, INTC, INTU, KLAC, LRCX, MCHP, META, MRVL, MSFT, MU, NFLX, NVDA, ORCL, PANW, QCOM, REGN, SNPS, TSLA, TXN, ISRG.

## 435 B.5 Compute

436 All experiments were run on a single workstation (Apple Silicon, 16 GB unified memory). LLM  
437 scoring:  $\approx 17.5$  hours for 40,850 articles at batch size 20 via a hosted chat-completions endpoint.  
438 DP-PPO training:  $\approx 45$  minutes per 30-epoch run (4 MPI workers, CPU-only PyTorch).

## 439 B.6 Drawdown shaping interpretation

440 Let  $D_t = \max(0, (W_{\text{peak}} - W_t)/W_{\text{peak}})$  denote fractional drawdown. The penalty  $-\alpha D_t^2$  is a smooth  
441 downside-risk surrogate: for small losses it is mild, but its gradient magnitude grows linearly in  
442 drawdown, so deeper excursions create increasingly strong pressure to reduce exposure. This differs  
443 from variance regularisation (symmetric upside/downside penalties) and from CVaR-style constraints  
444 (explicit tail optimisation). The intent is pragmatic shaping compatible with standard policy-gradient  
445 code.

## 446 C SAC mechanism hypotheses

447 The comparison suggests an optimisation explanation rather than a state-representation explanation  
448 alone: SAC exhibits lower realised volatility, smaller drawdown, shorter drawdown duration, and  
449 higher bear-market outperformance than DP-PPO under identical semantic inputs. A plausible  
450 mechanism is that on-policy PPO re-uses each transition only once: when only 14.8% of stock-days  
451 carry non-neutral semantic coordinates, PPO rollouts are dominated by neutral-signal steps, and  
452 rare informative transitions are weighted equally with uninformative ones. SAC’s experience replay  
453 allows semantic-rich transitions to be revisited proportionally more often during gradient updates,  
454 while entropy regularisation reduces premature policy collapse before semantic evidence accumulates.  
455 This “sparse-signal replay” effect is a property of the optimiser rather than the representation.

## 456 D Cross-domain SSAI instantiations

457 The SSAI framework (Definition 1) is not specific to portfolio trading *as a definition*. Any sequential  
458 system that occasionally observes sparse unstructured text *could* use a fixed-axis LLM interface as  
459 an auditable, sparse-by-design augmentation. **The present submission does not run experiments**  
460 **in these domains**; the following are *illustrative* sketches only: clinical triage (patient notes  $\rightarrow$   
461 urgency/confidence/risk); supply-chain optimisation (disruption news  $\rightarrow$  risk/confidence/lead-time);  
462 recommendation (reviews  $\rightarrow$  sentiment/engagement/quality). A diagnostic suite analogous to ours—  
463 direct factor tests, residualization, masking, optimiser comparison—*could* be instantiated elsewhere;  
464 doing so is future work. Prompts and the released harness are domain-agnostic at the software level,  
465 but all numbers in the main paper refer to equities only.

## 466 E Additional Results

### 467 E.1 Regime Analysis and Sub-period Breakdown

468 The 2019–2023 evaluation period spans three distinct regimes:

- 469 • **2019–2020 Q1**: bull market followed by COVID crash (S&P 500  $-34\%$  in 33 days,  
470 February–March 2020).
- 471 • **2020 Q2–2021**: recovery and growth rally.
- 472 • **2022**: Federal Reserve rate hike bear market (NASDAQ  $-33\%$  in 2022).

473 The turbulence index exceeded the threshold of 380 on 87 trading days (6.9% of the test period),  
474 all concentrated in March 2020 and 2022, triggering the no-buy constraint. Table 8 reports a more  
475 granular sub-period breakdown for the released checkpoint. DP-PPO outperforms buy-and-hold only  
476 during the 2020–2021 recovery bull window, while trailing during the pre-COVID, COVID-crash,  
477 2022 bear, and 2023 rally periods.

Table 8: Sub-period breakdown for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

| Period                            | Days | DP-PPO CR | B&H CR | $\Delta$ CR | Sharpe |
|-----------------------------------|------|-----------|--------|-------------|--------|
| Pre-COVID (2019–Feb 2020)         | 292  | 34.01     | 41.80  | -7.79       | 1.400  |
| COVID Crash (Mar–Apr 2020)        | 43   | -9.80     | 2.45   | -12.25      | -1.145 |
| Recovery Bull (May 2020–Dec 2021) | 422  | 144.86    | 110.56 | 34.30       | 2.159  |
| Rate Hike Bear (2022)             | 251  | -38.91    | -28.99 | -9.92       | -1.194 |
| 2023 Rally (2023)                 | 250  | 54.91     | 55.44  | -0.53       | 2.163  |

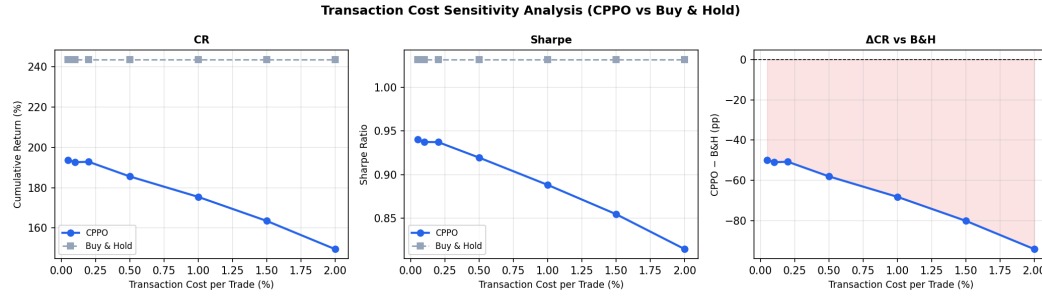


Figure 3: Transaction-cost sensitivity for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

## E.2 Transaction-Cost Sensitivity

Table 9 sweeps per-trade costs from 0.05% to 2.0% for the released DP-PPO checkpoint. Because buy-and-hold incurs only the initial allocation in this simplified sweep, DP-PPO remains below the passive sleeve at every tested cost level.

Table 9: Transaction-cost sensitivity for the released DP-PPO checkpoint. Buy-and-hold has no ongoing rebalance cost in this sweep; DP-PPO remains below the passive sleeve across tested cost levels.

| Cost (%) | DP-PPO CR | B&H CR | $\Delta$ CR | Sharpe | MDD (%) |
|----------|-----------|--------|-------------|--------|---------|
| 0.05     | 188.07    | 243.57 | -55.51      | 0.908  | -42.81  |
| 0.10     | 190.43    | 243.57 | -53.14      | 0.917  | -42.54  |
| 0.20     | 185.72    | 243.57 | -57.86      | 0.899  | -43.49  |
| 1.00     | 165.65    | 243.57 | -77.93      | 0.840  | -44.85  |
| 2.00     | 141.43    | 243.57 | -102.14     | 0.769  | -45.57  |

## E.3 Paired daily-return diagnostics (non-RL baselines)

## E.4 SFP Transaction-Cost Sensitivity

Table 11 sweeps per-trade costs from 0.05% to 2.0% for the four-factor SFP, mirroring Table 8 for DP-PPO. B&H uses the same equal-weight price-average index as Table 2 (243.6% CR); SFP uses the same 0.1%-cost evaluated portfolio as reported in the main results (307.2% CR at 0.1%). **Key finding:** SFP's daily top-10 rebalancing generates high turnover; at 0.1% (the baseline used in the main text) SFP exactly reproduces the reported 307.2% CR, but at 0.2% the gap over buy-and-hold reverses (214.5% vs. 243.6%), and at 0.5%+ SFP collapses. This means SFP's main-text outperformance is conditional on implementation at very low execution costs ( $\leq 0.1\%$ ), which practical portfolio managers should treat as a strong caveat.

Table 10: Paired daily-return diagnostics: semantic baselines plus multi-seed RL vs EW benchmark. The confidence interval is a 20-trading-day block bootstrap over mean paired daily active returns, in basis points per day. Wilcoxon tests are paired over daily returns and are reported as diagnostics rather than definitive multiple-comparison-adjusted claims.

| Comparison                                | Mean bp/day | 95% CI          | $\Delta$ Sharpe | Win % | Wilcoxon $p$ |
|-------------------------------------------|-------------|-----------------|-----------------|-------|--------------|
| SFP 4-factor vs sentiment-only            | 0.352       | [-1.070, 1.885] | +0.017          | 27.4  | 0.556        |
| SCW vs equal-weight SFP                   | 0.182       | [-1.224, 1.740] | +0.000          | 24.8  | 0.232        |
| Semantic tilt vs price-only forecaster    | 0.234       | [-1.198, 1.497] | +0.021          | 34.8  | 0.568        |
| Multi-seed DP-PPO mean vs EW buy-and-hold | -1.392      | [-4.119, 1.606] | -0.082          | 49.7  | 0.631        |

Table 11: SFP transaction-cost sensitivity (2019–2023). Each row re-evaluates the daily top-10 four-factor SFP at the stated per-trade cost. B&H (equal-weight price-average index, same computation as Table 2, 243.6% CR) is cost-free throughout. **SFP is highly turnover-sensitive**: it outperforms B&H only at  $\leq 0.1\%$  per trade and underperforms at  $\geq 0.2\%$ , collapsing at  $0.5\%+$ . The paper’s main reported result uses  $0.1\%$ ; realistic broker costs ( $0.2$ – $0.5\%$ ) reverse the advantage. Mirrors Table 8 (DP-PPO tx-cost sensitivity) for the direct portfolio.

| Cost (%) | SFP CR (%) | SFP Sharpe | SFP MDD (%) | B&H CR (%) | B&H Sharpe |
|----------|------------|------------|-------------|------------|------------|
| 0.05     | 363.3      | 1.151      | -35.6       | 243.6      | 1.032      |
| 0.10     | 307.2      | 1.067      | -35.6       | 243.6      | 1.032      |
| 0.20     | 214.5      | 0.899      | -35.6       | 243.6      | 1.032      |
| 0.50     | 44.7       | 0.395      | -59.3       | 243.6      | 1.032      |
| 1.00     | -60.5      | -0.436     | -86.1       | 243.6      | 1.032      |
| 2.00     | -97.1      | -2.002     | -98.9       | 243.6      | 1.032      |

## 492 E.5 SFP Sub-period Performance

493 Table 12 breaks down SFP performance by calendar year and regime window, mirroring Table 7 for  
494 DP-PPO. SFP outperforms buy-and-hold in 3 of 5 calendar years (2020: +11pp; 2021: +4pp; 2023:  
495 +26pp) and underperforms in 2019 (−8pp) and 2022 (−2pp). The largest single-year outperformance  
496 (+26pp in 2023) coincides with the AI-investment-cycle rally that disproportionately benefited  
497 NVDA, GOOGL, and AVGO—the high-coverage NASDAQ names that dominate the SFP basket (see  
498 SRF/SCW coincidence discussion in Section 5.2). **Caveat**: it is not possible to rule out that SFP’s  
499 2023 excess return is a disguised factor exposure to the AI-driven index rally rather than a semantic  
500 signal effect; a sector-neutral SFP design would be needed to isolate these contributions, which we  
501 leave as future work. Unlike DP-PPO—which outperforms only in the 2020–21 recovery—SFP’s  
502 positive excess return is concentrated in the 2020–21 recovery (+39pp) and 2023, both periods of  
high momentum in the NASDAQ-100 names that dominate the SFP selection basket.

Table 12: SFP sub-period performance (2019–2023). Mirrors Table 7 (DP-PPO sub-periods). Excess CR = SFP minus equal-weight B&H (same computation as Table 2). SFP outperforms in 3 of 5 calendar years (2020, 2021, 2023). The single largest annual outperformance (+26pp) occurs in 2023, coinciding with the AI investment cycle rally that disproportionately benefited the high-coverage NASDAQ names (NVDA, GOOGL, AVGO) that dominate the SFP basket; it is not possible to rule out that SFP’s 2023 excess is a disguised factor exposure rather than a semantic signal effect. SFP underperforms in 2019 and 2022 (rate-hike, drawdown year). Cost =  $0.1\%$  per trade.

| Period            | SFP CR (%) | SFP Sharpe | B&H CR (%) | Excess CR (pp) |
|-------------------|------------|------------|------------|----------------|
| 2019              | 35.6       | 1.460      | 43.3       | -7.7           |
| 2020              | 61.9       | 1.331      | 51.0       | +11.0          |
| 2021              | 46.4       | 1.827      | 42.3       | +4.1           |
| 2022              | -30.8      | -0.821     | -29.0      | -1.8           |
| 2023              | 81.5       | 2.857      | 55.4       | +26.1          |
| 2019–20 pre/bear  | 20.4       | 0.595      | 33.8       | -13.5          |
| 2020–21 recovery  | 179.0      | 2.334      | 139.8      | +39.2          |
| 2022–23 rate hike | 23.3       | 0.499      | 9.9        | +13.4          |

## 504 E.6 Coverage-Stratified SFP Analysis

505 Table 3 (reproduced in Section 5.2) addresses the coverage confound directly. We split the 30-ticker  
 506 universe into terciles by non-neutral coverage fraction and evaluate SFP within each tercile, comparing  
 507 against an equal-weight B&H sleeve of the *same tickers*. **Key result: SFP does not outperform**  
 508 **equal-weight B&H within any individual coverage tercile.** The full-portfolio outperformance  
 509 (307.2% vs. 243.6%) reflects a cross-universe *composition* effect—SFP’s top-10 rule preferentially  
 510 selects mid-coverage names (INTU, KLAC, LRCX, TSLA, ADBE) that happened to deliver strong  
 511 returns, not names on which the semantic signal is demonstrably more predictive. This result  
 512 substantially qualifies the main SFP finding; full discussion and the table appear in Section 5.2.

## 513 F LLM Signal Validation

Table 13: Descriptive statistics of LLM-generated signal scores (39,995 articles).

|        | Sentiment | Risk | Confidence | Vol. Forecast |
|--------|-----------|------|------------|---------------|
| Mean   | 3.35      | 2.47 | 3.51       | 2.74          |
| Std    | 1.01      | 1.02 | 0.77       | 0.82          |
| Median | 3.00      | 2.00 | 3.00       | 3.00          |

LLM Semantic Signal Validation

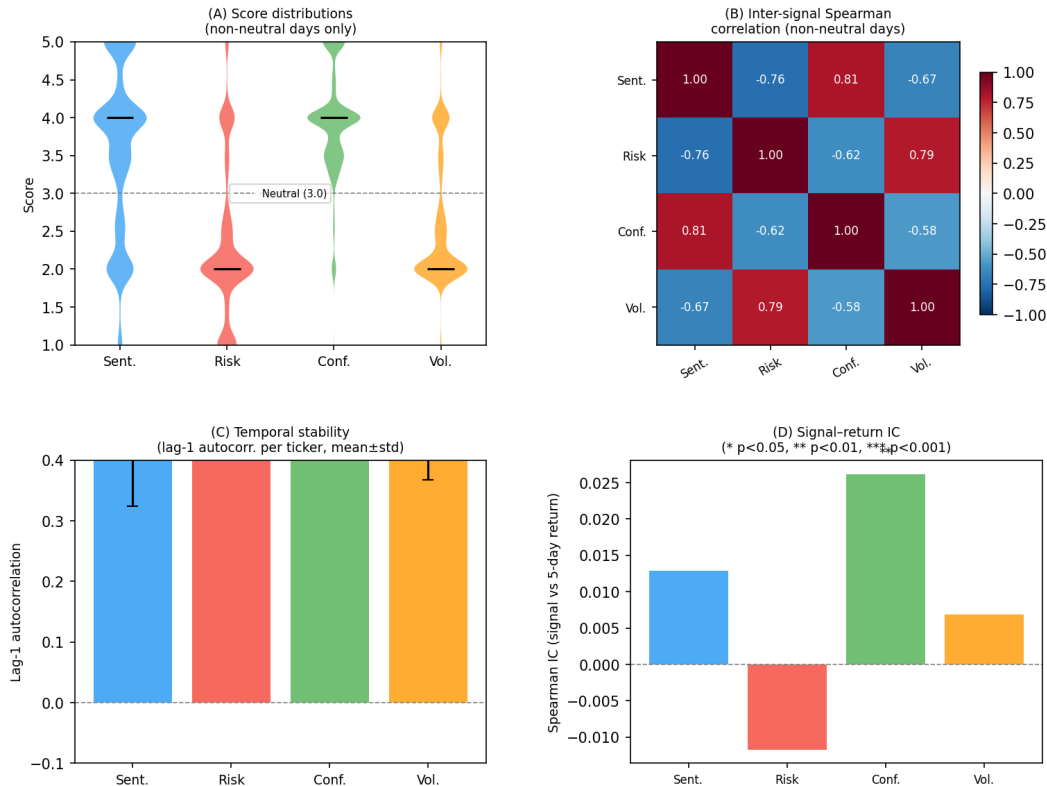


Figure 4: LLM semantic signal validation on the full 2013–2023 panel ( $N=83,040$  stock-days, 18,578 non-neutral). (A) Score distributions on non-neutral days show structured directional skew. (B) Inter-signal Spearman correlations replicate expected semantic structure. (C) Lag-1 temporal autocorrelation per ticker confirms scores are not white noise. (D) Signal–return IC: confidence shows a significant positive IC ( $p=0.004$ ).

Table 14: Ticker-level LLM signal coverage on the 2019–2023 trade panel. Coverage is the percentage of stock-days whose score differs from the neutral imputation value 3.0; rows show the five most-covered and five least-covered tickers.

| <b>Ticker</b> | <b>Stock-days</b> | <b>Any (%)</b> | <b>Sent.</b> | <b>Risk</b> | <b>Conf.</b> | <b>Vol.</b> |
|---------------|-------------------|----------------|--------------|-------------|--------------|-------------|
| GOOGL         | 1258              | 26.9           | 22.1         | 23.4        | 19.0         | 22.4        |
| AVGO          | 1258              | 26.6           | 19.8         | 24.4        | 16.6         | 23.1        |
| NVDA          | 1258              | 26.5           | 22.1         | 23.4        | 19.0         | 20.5        |
| MU            | 1258              | 24.6           | 21.2         | 22.8        | 18.0         | 21.4        |
| ASML          | 1258              | 23.4           | 17.9         | 19.5        | 13.3         | 16.1        |
| INTC          | 1258              | 0.5            | 0.5          | 0.5         | 0.5          | 0.5         |
| META          | 1258              | 0.0            | 0.0          | 0.0         | 0.0          | 0.0         |
| MSFT          | 1258              | 0.0            | 0.0          | 0.0         | 0.0          | 0.0         |
| ISRG          | 1258              | 0.0            | 0.0          | 0.0         | 0.0          | 0.0         |
| COST          | 1258              | 0.0            | 0.0          | 0.0         | 0.0          | 0.0         |

Table 15: Signal validity proxy: pooled Spearman information coefficients between LLM signal coordinates and 5-day forward returns / absolute returns on the 2019–2023 stock-date panel. This is not a human annotation audit, but it checks whether scores align with realized market quantities rather than pure noise.

| <b>Signal</b>       | <b>Return IC</b> | <b><i>p</i></b> | <b> Return  IC</b> | <b><i>p</i></b> |
|---------------------|------------------|-----------------|--------------------|-----------------|
| sentiment           | 0.0044           | 0.39            | -0.0047            | 0.36            |
| risk                | -0.0152          | 0.0032          | 0.0122             | 0.018           |
| confidence          | 0.0217           | 2.5e-05         | 0.0065             | 0.21            |
| volatility_forecast | -0.0045          | 0.38            | 0.0175             | 0.00068         |

## 514 G Ethics, Deployment, and Reproducibility

515 This work is a retrospective research benchmark, not a trading recommendation or deployable  
516 advisory system. Automated trading systems can amplify losses, liquidity stress, and feedback loops  
517 if deployed without governance, risk limits, and human oversight. Our evaluation omits important  
518 operational frictions—slippage, partial fills, exchange outages, short-sale constraints, tax effects,  
519 and capital limits—so reported results should not be interpreted as expected live performance. The  
520 LLM component introduces additional reproducibility concerns: hosted model behaviour can drift  
521 over time; prompts may elicit different outputs across providers or versions; and financial news  
522 coverage is uneven across firms. Scored article-level signals should be cached and released with  
523 prompt templates, model identifiers, timestamps, and dataset hashes. The artifact package provides  
524 scripts, processed CSV panels, checkpoints, and dry-run commands for retrained ablations, but full  
525 reproducibility still depends on access to the same market data snapshot and LLM-scored signal  
526 cache.

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